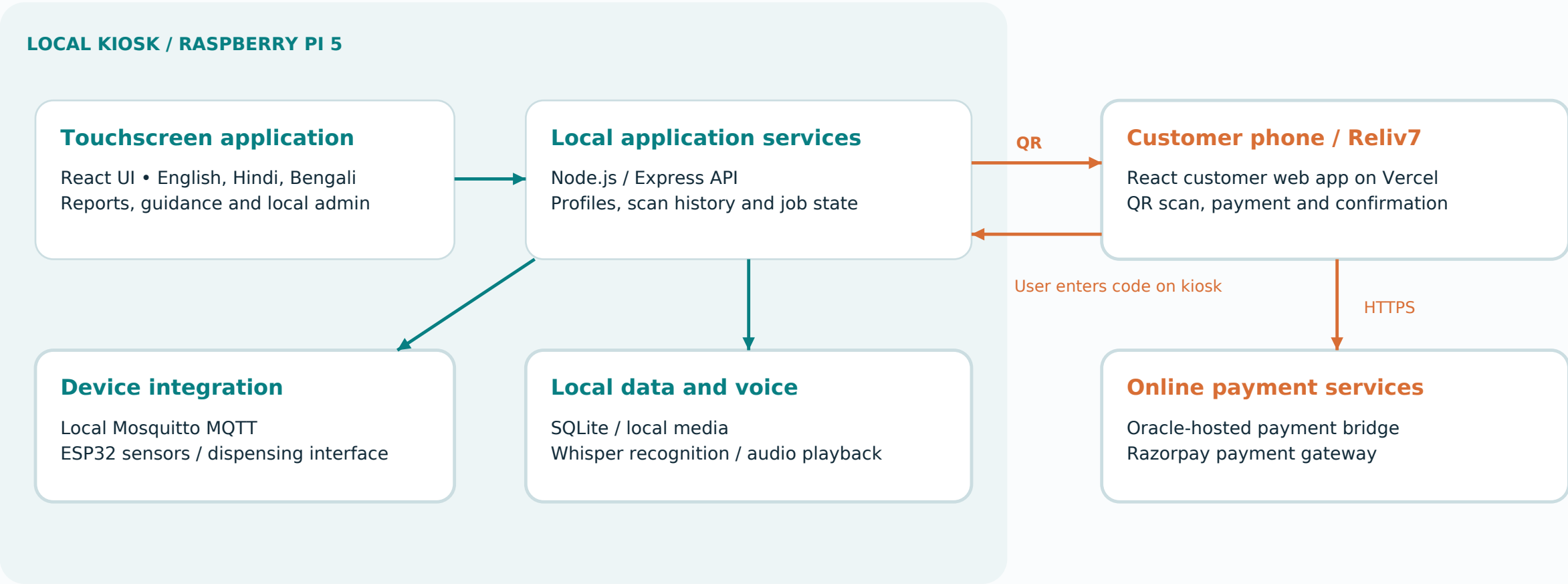


An offline kiosk. A separate online payment path.

Local sensing, records and kiosk operation stay on the Raspberry Pi. The customer phone provides the Internet connection.



No direct cloud-to-kiosk connection is required for this payment flow. Local ad uploads require a local connection; a fully disconnected kiosk cannot receive new Internet uploads automatically.

Technical stack

Core layers found in the repositories, with deployment context separated from source-code evidence.

Layer	Technology	Purpose / evidence
Kiosk interface	React 19 • Vite 7 • Tailwind CSS 3	Touchscreen flows, multilingual UI and report screens.
Customer web	React 18 • Vite 6 • Vercel	Reliv7 customer payment, QR scanner and shareable cards.
Backend / persistence	Node.js • Express 5 • SQLite / better-sqlite3	Local APIs, profiles, measurements, inventory and payment state.
Device communication	MQTT.js • Mosquitto • ESP32 / Arduino C++	Local commands and sensor results; sensor adapter firmware present.
Reports / visuals	Chart.js • PDFKit • jsPDF	Charts, derived metrics and document generation.
QR / card export	qrcode • qr-scanner • zxing-wasm html-to-image • html2canvas	QR creation / decoding and browser image export.
Language / voice	i18next • Python • whisper.cpp • eSpeak NG	UI localization, local recognition and speech fallback; bundled audio playback.
Deployment / payments	Raspberry Pi 5 • Linux • nginx Oracle bridge • Razorpay	Local deployment and cloud hosting per project setup; gateway SDK present.

Optional MongoDB, email and Drive integrations exist in the backend. They are not prerequisites for the intended offline kiosk operation.

AI / ML model and framework details

Speech recognition is the verified ML integration. Report personalization is implemented through application logic.

Capability	Implementation	Status / limits
Speech-to-text	Whisper multilingual base model via whisper.cpp. Documented model file: ggml-base-q5_1.bin.	Python microphone / WebSocket service calls a local recognizer. Model and binary must be installed on the Pi.
Languages / purpose	English, Hindi and Bengali recognition for guided kiosk interaction.	Transcription supports UI interaction; it does not itself authorize payments or dispensing.
Report personalization	Measured inputs, stored scan history, derived calculations and conditional narration.	No evidence found of a Reliv-trained diagnostic model or an LLM generating the report. Derived values must remain labelled as estimates.
Spoken output	Bundled recordings, browser speech support and local eSpeak NG fallback.	Separate from Whisper. Naturalness and available voices depend on installed assets and runtime configuration.
Audio asset production	edge_tts scripts generate some audio assets during preparation.	Online generation is distinct from offline playback. Voice / asset usage terms need separate review.

Suggested application-form wording

Reliv integrates local multilingual automatic speech recognition using Whisper through whisper.cpp. Health reports use deterministic calculations and personalized, rule-based explanations based on actual readings and scan history. No proprietary trained diagnostic AI model is claimed.

Open-source licence details

Selected direct dependencies and runtime components. This is a component summary, not a complete distribution licence audit.

Component group	Licence	Disclosure note
React, Vite, Tailwind CSS, Express, MQTT.js, Chart.js, PDFKit, qrcode, better-sqlite3	MIT	Licence metadata checked in repository lockfiles. Preserve applicable notices when distributing.
qr-scanner, html2canvas, html-to-image	MIT	Customer scanner and image-export dependencies.
zxing-wasm / underlying C++ barcode engine	MIT / Apache-2.0	Wrapper and native/WASM code have distinct licence coverage [S1].
Whisper code / model weights; whisper.cpp	MIT	Local recognition integration; verify the exact deployed model and build [S2, S3].
SQLite core	Public domain	Distinct from the MIT-licensed Node.js wrapper [S4].
Eclipse Mosquitto	EPL-2.0 OR EDL-1.0	Dual-licensed broker; shipped version and notices must be recorded [S5].
eSpeak NG	GPL-3.0-or-later	Local speech executable; review relevant redistribution obligations [S6].
Lucide icons / bundled Noto fonts	ISC / SIL OFL-1.1	Lockfile icon metadata and bundled font licence files.

Reliv application ownership is separate from third-party component licences. No top-level Reliv licence was established in this review. OS packages, firmware libraries, model binaries, media and service terms require a final shipped-component inventory.

Evidence and disclosure scope

A source-backed technical overview suitable for an initial submission, without confidential implementation details.

What was inspected

Frontend manifests / lockfiles, report and voice modules, audio-generation scripts; backend API / service files, body-estimate modules, voice-service documentation and client code, plus ESP32 sensor-adapter firmware.

What this document does not certify

The physical Pi deployment, sensor accuracy, dispensing hardware, installed voice quality, real payment settlement or complete licence compliance. No deployment or functional test certification is implied by a documentation review.

Primary licence references

S1 ZXing WASM licensing

<https://github.com/Sec-ant/zxing-wasm#license>

S2 Whisper code and model licences

<https://github.com/openai/whisper#license>

S3 whisper.cpp licence

<https://github.com/ggml-org/whisper.cpp/blob/master/LICENSE>

S4 SQLite copyright / public domain

<https://www.sqlite.org/copyright.html>

S5 Eclipse Mosquitto licence

<https://github.com/eclipse-mosquitto/mosquitto/blob/master/LICENSE.txt>

S6 eSpeak NG licence information

<https://github.com/espeak-ng/espeak-ng#license-information>

Review record

Backend source snapshot: 9bf530a. Frontend base: 8275346 with the reviewed scanner / export working changes. This overview omits credentials, network identifiers, signing details, scoring formulas, wiring and cabinet dimensions. Architectural facts alone do not establish patentability or ownership.